

A Comparative Analysis of Compute Infrastructure Services Across Amazon Web Services, Microsoft Azure, and Google Cloud Platform

Lathika Busetty

Faculty: Rama Tulasi

Department of Computer Science and Engineering
VIT-AP University, India

Abstract—Cloud computing has become an important foundation for modern application development, digital transformation, and scalable infrastructure. Among the major cloud service providers, Amazon Web Services (AWS), Microsoft Azure, and Google Cloud Platform (GCP) provide extensive compute capabilities through virtual machines, container services, serverless computing, and automated scaling mechanisms.

This paper presents a comparative analysis of AWS, Azure, and GCP with particular emphasis on compute infrastructure. The comparison considers virtual machine services, container orchestration, serverless computing, performance and scalability, pricing models, regional availability, networking capabilities, security mechanisms, compliance frameworks, DevOps integration, and hybrid and edge computing capabilities. The study also examines representative real-world use cases associated with each platform.

The analysis indicates that the three providers offer comparable core compute capabilities while differentiating themselves through their infrastructure technologies, ecosystem integration, pricing mechanisms, hybrid-cloud capabilities, Kubernetes support, and global infrastructure. The study provides a structured basis for understanding the characteristics of the three platforms and selecting compute infrastructure according to technical requirements, workload characteristics, scalability needs, security requirements, and cost considerations.

Index Terms—Cloud Computing, AWS, Microsoft Azure, Google Cloud Platform, Virtual Machines, Containers, Kubernetes, Serverless Computing, Auto Scaling, Cloud Security, Cloud Pricing

I. INTRODUCTION

Cloud computing has transformed the way organizations develop, deploy, and operate software applications. Instead of maintaining physical infrastructure, organizations can obtain computing resources on demand through cloud service providers. Compute infrastructure is particularly important because application performance, availability, scalability, and operational cost are strongly influenced by the underlying compute platform.

Amazon Web Services (AWS), Microsoft Azure, and Google Cloud Platform (GCP) are major cloud platforms offering a broad collection of compute services. Although their fundamental objectives are similar, each provider offers different technologies, service configurations, pricing approaches, security mechanisms, and ecosystem integrations.

This research performs a comparative analysis of the compute infrastructure systems of AWS, Azure, and GCP. The

study examines virtual machines, container-based services, serverless computing, scalability, performance, pricing, regional availability, networking, security, DevOps, and hybrid and edge computing.

The comparison is useful for organizations and developers who need to select a cloud platform for applications such as scalable web systems, microservices, high-performance computing (HPC), and containerized CI/CD environments.

II. SIGNIFICANCE OF THE STUDY

The expansion of cloud computing has increased the need for organizations to identify appropriate infrastructure for application deployment. The compute layer influences application performance, availability, user experience, and infrastructure cost.

AWS provides technologies such as the Nitro System together with a broad collection of cloud services. Azure provides strong hybrid-cloud capabilities and integration with enterprise and Microsoft-based workloads. GCP provides extensive Kubernetes capabilities, advanced networking, and strong support for data-intensive workloads.

Therefore, cloud provider selection requires more than comparing service names. Important factors include:

- Virtual machine capabilities and configurations.
- Container orchestration and serverless container services.
- Serverless execution capabilities.
- Automatic scaling mechanisms.
- Pricing and cost optimization options.
- Regional and availability-zone coverage.
- Network performance.
- Security and compliance mechanisms.
- DevOps and CI/CD integration.
- Hybrid and edge computing capabilities.

III. SCOPE OF THE ANALYSIS

The study focuses on compute-specific services and supporting infrastructure components. The major areas covered are:

- Virtual Machines (IaaS)
- Container Services
- Serverless Computing (FaaS)
- Performance and scalability

- CPU, memory, disk, and network considerations
- Cost models and optimization
- Regional availability and edge presence
- Security mechanisms
- Compliance frameworks
- DevOps and CI/CD integration
- Hybrid and edge computing

Storage, artificial intelligence, and database services are outside the primary scope unless they directly support compute operations.

IV. RESEARCH METHODOLOGY

The study uses a comparative research methodology based on official documentation, technical white papers, benchmarking reports, pricing information, and practical use cases.

The research process consists of the following stages:

- 1) Identification of the major compute services provided by AWS, Azure, and GCP.
- 2) Comparison of virtual machine families and configuration capabilities.
- 3) Examination of container orchestration and serverless container services.
- 4) Comparison of serverless function platforms.
- 5) Analysis of performance and automatic scaling mechanisms.
- 6) Examination of pricing and cost optimization models.
- 7) Comparison of identity management and compliance mechanisms.
- 8) Analysis of DevOps, hybrid-cloud, and edge computing support.
- 9) Examination of representative real-world use cases.

Public documentation and pricing resources from the three cloud providers form an important part of the study. The original study also considers benchmarking reports and practical case studies to supplement the technical comparison.

V. COMPUTE SERVICES COMPARISON

A. Virtual Machines

Virtual machines are one of the fundamental compute services offered by cloud providers. They provide configurable computing environments where users can select CPU, memory, storage, networking, and accelerator resources according to workload requirements.

1) *Amazon EC2*: Amazon Elastic Compute Cloud (EC2) provides multiple instance families:

- **General Purpose**: Balanced CPU and memory resources.
- **Compute Optimized**: Designed for compute-intensive workloads.
- **Memory Optimized**: Suitable for memory-intensive applications.
- **Storage Optimized**: Designed for workloads requiring high read and write performance.

EC2 also supports GPU-enabled instances and allows users to select specific CPU, memory, storage, and networking configurations.

AWS provides several pricing models including On-Demand pricing, Savings Plans, Reserved Instances, and Spot Instances. Spot Instances provide access to unused computing capacity at reduced prices and are particularly suitable for fault-tolerant workloads.

2) *Microsoft Azure Virtual Machines*: Azure provides several VM series designed for different workloads:

- **A-Series**: Entry-level machines for development and testing.
- **D-Series**: General-purpose workloads.
- **E-Series**: Memory-intensive applications.
- **F-Series**: Compute-intensive workloads.
- **G-Series**: High-memory and storage workloads.
- **H-Series**: High-performance computing.

Azure also provides Accelerated Networking to improve network throughput and reduce latency.

Its pricing models include Pay-As-You-Go, Reserved VM Instances, and Spot VMs.

3) *Google Compute Engine*: Google Compute Engine provides several machine families:

- General-purpose machines.
- Compute-optimized machines.
- Memory-optimized machines.
- Accelerator-optimized machines.

GCP also provides customizable machine types that allow users to select CPU and memory configurations according to workload requirements.

Pricing options include pay-as-you-go pricing, committed-use discounts, and Spot VMs.

B. Container Services

Container technologies provide lightweight application packaging and support modern microservices architectures.

1) *AWS*: AWS provides:

- **Amazon ECS**: A scalable container orchestration service.
- **Amazon EKS**: A managed Kubernetes service.
- **AWS Fargate**: A serverless compute engine for containers.

ECS integrates with EC2 and Fargate, while EKS provides managed Kubernetes capabilities. Fargate allows users to execute containers without directly managing the underlying infrastructure.

2) *Azure*: Azure provides:

- **Azure Kubernetes Service (AKS)**: Managed Kubernetes.
- **Azure Container Instances (ACI)**: Serverless container execution.
- **Azure Container Registry**: Private container image storage.

AKS integrates with Azure Monitor, Azure Active Directory, and Azure DevOps and supports both Linux and Windows containers.

TABLE I
COMPARISON OF VIRTUAL MACHINE SERVICES

Feature	AWS EC2	Microsoft Azure VMs	Google Compute Engine
General Purpose	Balanced resources for different workloads	D-Series and related general-purpose configurations	General-purpose machine families
Compute Optimized	Compute-intensive workloads	F-Series	Compute-optimized machine families
Memory Optimized Accelerators	Memory-intensive applications GPU-enabled instances	E-Series GPU-enabled configurations	Memory-optimized machines GPU and TPU-enabled configurations
Customization	CPU, memory, storage and networking options	Multiple VM sizes and configurations	Custom machine types
Discount Models	Savings Plans, Reserved Instances, Spot Instances	Reserved Instances and Spot VMs	Committed Use Discounts and Spot VMs

3) *GCP*: Google Cloud provides:

- **Google Kubernetes Engine (GKE)**: Managed Kubernetes.
- **Cloud Run**: Serverless container execution.
- **Artifact Registry**: Storage for container images and software packages.

GKE provides both Autopilot and Standard operating modes, while Cloud Run provides automatic scaling and can scale services to zero when they are not receiving requests.

C. Serverless Computing

Serverless computing allows developers to execute application logic without directly managing servers.

1) *AWS Lambda*: AWS Lambda provides event-driven execution and integrates with services such as API Gateway, Amazon S3, and DynamoDB.

Lambda supports languages including:

- Node.js
- Python
- Java
- Go
- Ruby
- C#

The study identifies a maximum execution duration of 15 minutes. Lambda automatically scales according to incoming requests and provides integration with CloudWatch and X-Ray.

2) *Azure Functions*: Azure Functions provides an event-driven serverless environment supporting HTTP triggers, timer triggers, Event Grid, Service Bus, and storage-based events.

It supports C#, JavaScript, Python, PowerShell, Java, and other languages. Durable Functions provide workflow capabilities including function chaining and fan-out/fan-in patterns.

The study identifies different execution characteristics for consumption, premium, and dedicated plans.

3) *Google Cloud Functions*: Google Cloud Functions provides event-driven execution for HTTP and background workloads. It integrates with services such as Pub/Sub, Firestore, Firebase, and Cloud Storage.

The study describes first-generation and second-generation execution models, with second-generation functions providing longer execution duration and improved concurrency control.

VI. PERFORMANCE AND SCALABILITY

Cloud provider selection requires consideration of performance and scalability because applications may require continuous availability, rapid response, and automatic resource management.

The major benchmarking considerations include:

A. VM Boot Time

VM boot time represents the time required for a virtual machine to move from resource allocation to an operational state. Lower boot times can support faster response during automatic scaling.

B. CPU and Memory Throughput

CPU and memory throughput indicate the ability of VM instances to process demanding workloads. Hardware specifications, cloud optimizations, and the selected instance type influence these characteristics.

C. Disk I/O

Disk I/O latency and throughput determine how quickly a system can read from and write to connected storage volumes. Low latency and high throughput can be important for databases, analytics, and media processing workloads.

D. Automatic Scaling

Automatic scaling allows cloud resources to adjust according to workload requirements.

1) *AWS Auto Scaling*: AWS provides Auto Scaling Groups (ASGs) for managing EC2 instance numbers. Elastic Load Balancing can distribute traffic across instances.

AWS supports:

- Target tracking scaling.
- Step scaling.
- Scheduled scaling.

2) *Azure Auto Scaling*: Azure Virtual Machine Scale Sets (VMSS) manage groups of virtual machines. Azure Monitor can be used to observe performance metrics.

Azure supports manual and rule-based scaling, including horizontal and vertical scaling.

TABLE II
SERVERLESS COMPUTING COMPARISON

Feature	AWS Lambda	Azure Functions	Google Cloud Functions
Languages	Node.js, Python, Java, Go, Ruby, C#	C#, JavaScript, Python, PowerShell, Java	Node.js, Python, Go, Java, .NET, Ruby, PHP
Maximum Timeout	15 minutes	5 minutes on consumption plan; higher limits on other plans	9 minutes for first generation; up to 60 minutes for second generation
Pricing	Pay-as-you-go and GB-second usage	Consumption, Premium and Dedicated options	Pay-as-you-go based on invocation, compute and memory
Cold Start	Present; mitigation mechanisms available	Present; reduced in premium configurations	Present; improved in second generation

3) *GCP Auto Scaling*: Google Cloud uses Managed Instance Groups (MIGs) to manage groups of virtual machines. Autoscaling can use CPU utilization, load balancing, and user-defined metrics.

Instance templates help maintain consistent VM configurations.

E. Network Performance

Network infrastructure strongly influences application performance and data transmission speed.

1) *AWS Nitro System*: The AWS Nitro System combines hardware and virtualization technologies to improve networking, security, and I/O performance. The study identifies network capabilities reaching 100 Gbps for specific instance types.

2) *Azure Accelerated Networking*: Azure Accelerated Networking uses Single Root I/O Virtualization (SR-IOV) to allow network packets to bypass portions of the host virtualization stack. This can reduce CPU utilization, network jitter, and latency.

3) *Google Cloud Premium Tier*: Google Cloud Premium Tier uses Google's global private fiber network. It provides global load balancing and high-performance network connectivity.

VII. PRICING MODELS AND COST OPTIMIZATION

AWS, Azure, and GCP provide multiple pricing approaches for different workload patterns.

A. Pay-As-You-Go

Pay-As-You-Go pricing allows customers to pay according to actual resource usage without long-term commitments. This model is suitable for short-term or variable workloads.

B. Reserved and Commitment Discounts

Long-term commitments can provide discounts for predictable workloads. AWS provides Savings Plans and Reserved Instances, Azure provides Reserved VM Instances, and GCP provides Committed Use Discounts.

C. Spot and Preemptible Resources

Cloud providers offer discounted access to unused compute resources. AWS provides Spot Instances, Azure provides Spot VMs, and GCP provides Spot VMs.

These resources can be useful for workloads that tolerate interruption, such as batch processing and fault-tolerant applications.

D. Free Tiers

The study identifies free-tier or trial resources across all three providers. These offerings can allow students, developers, and small projects to experiment with cloud infrastructure before moving to paid resources.

VIII. SECURITY AND COMPLIANCE

Security is an important component of cloud infrastructure. AWS, Azure, and GCP provide identity management, role-based access control, monitoring, and compliance mechanisms.

A. AWS IAM

AWS Identity and Access Management (IAM) allows users to define permissions through policies, roles, and groups. It supports multi-factor authentication and federated identity environments.

B. Azure RBAC

Azure Role-Based Access Control (RBAC) provides authorization through role assignments at subscription, resource-group, and resource levels. Both built-in and custom roles are supported.

C. Google Cloud IAM

Google Cloud IAM provides fine-grained access control across projects, folders, and organizations. It supports primitive, predefined, and custom roles.

TABLE III
COMPLIANCE FRAMEWORKS

Standard	AWS	Azure	GCP
ISO/IEC 27001	Yes	Yes	Yes
ISO/IEC 27017	Yes	Yes	Yes
ISO/IEC 27018	Yes	Yes	Yes
SOC	Yes	Yes	Yes
GDPR	Yes	Yes	Yes
HIPAA	Yes	Yes	Yes
FedRAMP	Yes	Yes	Yes
PCI DSS	Yes	Yes	Yes

The study also identifies provider-specific compliance portals: AWS Artifact, Azure Trust Center, and the Google Cloud Compliance Resource Center.

IX. ECOSYSTEM INTEGRATION

A. DevOps and CI/CD

Cloud platforms provide services supporting continuous integration, continuous delivery, automation, and Infrastructure as Code.

1) *AWS*: AWS provides:

- CodePipeline for CI/CD workflows.
- CodeBuild for building and testing software.
- CodeDeploy for automated deployment.

2) *Azure*: Azure DevOps integrates repositories, pipelines, boards, and artifacts. GitHub Actions provides event-driven CI/CD workflows, while Bicep supports Infrastructure as Code for Azure resources.

3) *GCP*: Google Cloud provides Cloud Build for serverless CI/CD and Cloud Deploy for deployment strategies such as canary and blue/green deployments. Terraform and other infrastructure automation tools can also be used.

B. Hybrid and Edge Computing

Cloud providers increasingly support hybrid and edge environments.

1) *AWS*: AWS Outposts extends AWS infrastructure to customer facilities. Snowball and Snowcone provide portable computing and data-transfer capabilities for remote environments.

2) *Azure*: Azure Arc enables management of resources outside Azure, including on-premises and multi-cloud environments. Azure Stack provides Azure services on customer-managed infrastructure.

3) *GCP*: Google Anthos provides capabilities for hybrid and multi-cloud Kubernetes environments. Edge technologies are also supported for computing at distributed locations.

X. USE CASES AND REAL-WORLD EXAMPLES

A. AWS

The study identifies several organizations using AWS infrastructure.

Netflix: AWS supports global content delivery and scalable application infrastructure.

Airbnb: AWS infrastructure is used for compute, storage, and content delivery.

NASA: AWS has been used for handling large volumes of satellite data and supporting scientific and climate-related workloads.

B. Microsoft Azure

BMW: Azure services are used for processing data from manufacturing environments.

Adobe: Azure infrastructure supports Adobe cloud services and enterprise workloads.

HP: Azure is used for enterprise workloads and customer-service applications.

C. Google Cloud Platform

Spotify: GCP services support recommendation-related workloads and large-scale data processing.

Snapshot: GCP services are used for image processing and scalable application infrastructure.

PayPal: GCP infrastructure has been used for AI-assisted fraud detection and secure payment-related workloads.

XI. COMPARATIVE SUMMARY

Table IV summarizes the major compute services discussed in the study.

TABLE IV
AWS, AZURE AND GCP COMPUTE COMPARISON

Feature	AWS	Azure	GCP
VM Service	EC2	Azure VMs	Compute Engine
Kubernetes	EKS	AKS	GKE
Serverless	Lambda	Azure Functions	Cloud Functions
Auto Scaling	ASG	VMSS	MIG
Pricing Tools	Cost Explorer	Cost Management	Billing Reports
Compliance	SOC, ISO, HIPAA, GDPR	SOC, ISO, HIPAA, GDPR	SOC, ISO, HIPAA, GDPR

XII. DISCUSSION

The comparison demonstrates that AWS, Azure, and GCP provide a broad range of compute services but emphasize different aspects of cloud infrastructure.

AWS provides a mature compute ecosystem through EC2, ECS, EKS, Fargate, and Lambda. Its infrastructure technologies, pricing models, and broad service integration provide multiple deployment options.

Azure provides strong integration with enterprise infrastructure, Microsoft technologies, Azure DevOps, Windows workloads, and hybrid cloud environments. Its VM, Kubernetes, and serverless services provide multiple options for enterprise application deployment.

GCP provides extensive Kubernetes capabilities through GKE and serverless container support through Cloud Run. Its networking infrastructure and machine configurations support applications requiring scalable compute and distributed workloads.

The appropriate choice therefore depends on application requirements rather than a single universal platform. Workload characteristics, expected traffic, compute requirements, security requirements, geographic requirements, existing enterprise infrastructure, and cost models should all be considered.

XIII. CHALLENGES IN CLOUD PROVIDER SELECTION

Several challenges can affect cloud provider selection:

- Rapid changes in cloud service configurations.
- Complex pricing structures.
- Differences in regional service availability.
- Vendor-specific APIs and management tools.
- Potential vendor lock-in.
- Differences in performance between VM configurations.
- Complexity of multi-cloud and hybrid-cloud management.
- Need for continuous monitoring and cost optimization.

These factors make direct comparison difficult when workloads differ significantly.

XIV. FUTURE SCOPE

Future studies can extend this research by conducting controlled performance experiments across equivalent VM configurations.

Potential areas include:

- Automated VM boot-time benchmarking.
- CPU and memory performance experiments.
- Network latency and throughput measurements.
- Container startup-time comparisons.
- Serverless cold-start experiments.
- Detailed cost analysis for different workload patterns.
- Energy efficiency and sustainability comparison.
- Multi-cloud deployment experiments.
- Edge computing performance evaluation.

Such experiments could complement the documentation-based comparison presented in this study.

XV. CONCLUSION

This paper presented a comparative analysis of the compute infrastructure provided by Amazon Web Services, Microsoft Azure, and Google Cloud Platform.

The comparison covered virtual machines, containers, serverless computing, scalability, networking, pricing, security, compliance, DevOps, hybrid cloud, edge computing, and practical use cases.

AWS provides a broad and mature compute ecosystem with services such as EC2, ECS, EKS, Fargate, and Lambda. Azure provides extensive enterprise and hybrid-cloud capabilities through Azure Virtual Machines, AKS, Azure Functions, and Azure DevOps. GCP provides strong Kubernetes and serverless container capabilities through GKE and Cloud Run, together with its global networking infrastructure.

The study shows that cloud provider selection should be based on the specific requirements of an application. Factors such as performance, scalability, pricing, security, geographic

availability, ecosystem integration, and workload characteristics should be evaluated before deployment.

Therefore, rather than considering cloud computing platforms as interchangeable services, organizations should analyze their workload requirements and match those requirements with the capabilities offered by each provider.

REFERENCES

- [1] Amazon Web Services, "AWS Documentation," Available: <https://docs.aws.amazon.com/>
- [2] Microsoft, "Microsoft Azure Documentation," Available: <https://learn.microsoft.com/en-us/azure/>
- [3] Google Cloud, "Google Cloud Documentation," Available: <https://cloud.google.com/docs>
- [4] Cloud Spectator, "Cloud Performance Benchmarks," Available: <https://cloudspectator.com/reports/>
- [5] Phoronix, "Cloud Benchmarks," Available: <https://www.phoronix.com/scan.php?page=home>
- [6] Cockroach Labs, "The Cloud Report," Available: <https://www.cockroachlabs.com/blog/2023-cloud-report/>
- [7] Gartner, "Magic Quadrant for Cloud Infrastructure and Platform Services," Available: <https://www.gartner.com/en/research/magic-quadrant>
- [8] Forrester, "Forrester Wave Research," Available: <https://go.forrester.com/research/>
- [9] IDC, "IDC MarketScape Reports," Available: <https://www.idc.com/>
- [10] Amazon Web Services, "AWS Whitepapers Library," Available: <https://aws.amazon.com/whitepapers/>
- [11] Microsoft, "Azure Architecture Center," Available: <https://learn.microsoft.com/en-us/azure/architecture/>
- [12] Google Cloud, "Google Cloud Solutions Library," Available: <https://cloud.google.com/solutions>
- [13] HashiCorp, "Infrastructure and Multi-Cloud Resources," Available: <https://www.hashicorp.com/resources>
- [14] IEEE, "IEEE Xplore Digital Library," Available: <https://ieeexplore.ieee.org/>
- [15] Elsevier, "ScienceDirect," Available: <https://www.sciencedirect.com/>
- [16] Springer, "SpringerLink," Available: <https://link.springer.com/>
- [17] StackShare, "Cloud Technology Comparisons," Available: <https://stackshare.io/>
- [18] CloudHarmony, "Cloud Performance and Availability Data," Available: <https://cloudharmony.com/>